



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EM/LMI/566909

Date:

13/10/2023(17:23:48)

Subject:Revision of LMI Prevention and detection of generator rotor earth faults
LMI/OGN/OPS/ELEC/007

LMIs are reviewed periodically for further improvement and incorporation of recommendation by internal or external audit committee and COS.No changes envisaged by COS and OEM.

Minor changes were made inline with COS document and revision document "LMI on Generator rotor earth faults_revised.pdf" is put up for review.

Submitted for review & approval of competent authority please.

Submitted by:

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Note No:2

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Note No:3

LMI reviewed and found in order

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Note No:4

Review as discussed

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Note No:5

Sir,As discussed reviewed and modified.

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Note No:6

LMI reviewed , revised and check list incorporated

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Note No:7

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Note No:9

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On:04/11/2023(18:59:26) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM/LMI/566909

Note No:10

Proposed LMI is reviewed and found in line with OGN.

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On:06/11/2023(18:16:39) **Ref:**SMTTPP/ELECT MAINT/2023-24/EM/LMI/566909

Note No:11

Please consider the following changes:-

- Change the sheet no.1 where it is mentioned "IMPLEMENTATION BY
" (to) - by BUH (Simhadri) or HOP(Simhadri)
- On page no./sheet no. 7 of the revised LMI in clause no. 12. Please consider replacing the old /existing statement with the following statement “Basic Responsibility lies with respective Area In-charge i.e. (EMD-Lab). Overall responsibility lies with HOD (EMD)- I/c.”.

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Note No:12

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Note No:13

Required changes as mentioned in earlier note pages was done and new file attached as LMI Prevention and detection of generator rotor earth faults_181123.pdf.

Put up for review and approval please.

Submitted by:

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Note No:14

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Note No:15

Modified and Reviewed

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Note No:16

Revised as per the observation made by GM- Maintenace, may be approved.

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Note No:17

May please be accorded approval.

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Note No:18

Approved & Submitted by:

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**NTPC LIMITED
SIMHADRI THERMAL POWER STATION**

LOCAL MANAGEMENT INSTRUCTION
LMI/COS-ISO-00-OGN/OPS/ELEC/007
ISSUE NO.: 1, REVISION : 4, DATE: OCT 2023

**Prevention and Detection of Generator
Rotor Earth Faults**

APPROVED FOR

IMPLEMENTATION BY_-----
HOP (Simhadri)

DATE -----

NTPC LTD., SIMHADRI THERMAL POWER STATION
OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	Prevention and Detection of Generator Rotor Earth Faults		
LMI No.	LMI/COS-ISO-00-OGN/OPS/ELEC/007		
ISSUE NO.	1	Revision : 4	Date :OCT 2023

	DESIGNATION	SIGNATURE AND NAME
Prepared by	Sr Manager (EM)	
Checked by	AGM (EM)	
Checked by	HOD (EM)	
Checked by	GM(Maint)	
Checked by	GM(Operation)	
Reviewed by	GM(O&M)	

Prevention and Detection of Generator Rotor Earth Faults

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1. INTRODUCTION :

Rotor Earth Fault on Generator is responsible for many forced outages and shutdown of machines. If the generator is allowed to run with one rotor earth fault, no damage to the rotor is apprehended because the return path of the current is not available. However, if second rotor earth fault gets developed in the process, it could result in serious and extended damage to rotor steel forging and winding and subsequent damage to the seals. The magnetization of the rotor shaft is quite common on account of the rotor earth faults and can also result in permanent damage. Immediate change noticed in the operating parameters will be, the increase in field current and generator bearing vibrations.

The objective of this LMI is to describe the causes and detections of rotor earth fault.

2. SCOPE :

This document covers the generator rotor of Simhadri Unit 1, 2, 3 & 4.

3. REFERENCE DOCUMENTS :

COS-ISO-00-OGN/OPS/ELEC/007.

4. SUPERSEDED DOCUMENTS :

LMI/COS-ISO-00-OGN/OPS/ELEC/007 ISSUE NO.: 1, REV : 3, DT: Jan 2022

Additions:

1. Damage on account of double rotor earth faults
2. Philosophy of second stage rotor earth fault
3. Rotor earth fault detection
4. Checklist for prevention and detection of generator rotor earth faults

5. Possible Causes for Rotor earth Fault:

A. Temporary faults creating creepage to ground but not causing longer forced unit outages.

- a) Moisture
- b) Oil
- c) Dust particles

B. Permanent faults could be due to :

- a) Insulation failure between rotor body and rotor copper conductor in slot

portion or leads

- b) Current carrying bolts insulation failure.
- c) Main exciter rotor winding insulation failure.
- d) DC leads through multi contact pins insulation failure.
- e) Diode wheel assembly insulation failure.

The moisture enters the generator if hydrogen dryers are not effective. Oil enters into the Generator due to problem in the seals and subsequent oil spray. Dust particles find their way into the rotor ventilating canals when rotors are not covered properly during maintenance after threading out or during erection/storage stage itself. Hence there are chances of dirt getting into the coolant stream and thereby getting deposited at critical areas to create creepage shorts. Due care during storage, erection, operation and maintenance of these equipments will prevent rotor earth fault and subsequent rotor damage.

6. DAMAGE ON ACCOUNT OF DOUBLE ROTOR EARTH FAULTS:

If the generator is allowed to run with one rotor earth fault, no damage to the rotor is apprehended because the return path of the current is not available. However, if second rotor earth fault gets developed in the process, it could result in serious and extended damage to rotor steel forging and windings and subsequent damage to the seals. The magnetization of the rotor shafts is quite common on account of the rotor earth faults and can also result in permanent damage. Immediate change noticed in the operating parameters will be the increase in the field current and generator bearing vibrations.

7. PHILOSOPHY OF SECOND STAGE ROTOR EARTH FAULT:

The philosophy of unit tripping on second stage rotor earth fault had also been under discussion in the past. The practice for tripping of the units on rotor earth faults varies from manufacturer to manufacturer. The second stage protection in fact does not allow the second rotor earth fault but is the extension of the first fault itself. The Insulation Resistance (IR) of the rotor is monitored and if it falls below a

predetermined value, alarm will appear and if decay of the IR value continues, the unit is allowed to trip. However, appearance of the rotor earth fault alarm should be the cause of concern for the operating staff and rectification measures are needed immediately. Whenever Rotor Earth fault alarm appears, the Shift Engineer should inform the same to Electrical Maintenance In-charge for taking necessary remedial measures.

8. ROTOR EARTH FAULT DETECTION :

A D.C. voltage is applied between the rotor shaft so that around 500-700 Amp current flows through the shaft. One end of the galvanometer is connected with one slip ring and the other end moved along the rotor shaft. The null position of the galvanometer will indicate the location of the fault in that plane. After determining the plane of the fault, remove the D.C. source from rotor ends and apply D.C voltage with high current across the plane. The moveable contact of the galvanometer is moved around the rotor barrel periphery. The point where meter shows zero deflection will indicate the exact location of the fault.

9. SPECIFIC INSTRUCTIONS TO BE FOLLOWED:

Instruction	Responsibility	Document Ref.
Inform to Electrical Maint. in case Rotor Earth Fault alarm appears.	Operation/SCE	DDCMIS alarm history (kks code-00MKY01GX022)
To ascertain actual cause of alarm and recommend immediate action.	Area in-charge of Generator Maintenance group	1. Operation shift log book (TG desk engineer) 2. File:LMI/E-007
Testing of rotor earth fault protection during unit overhaul.	Area in-charge of Electrical testing.	File : 1. OH Report 2. Relay test results.

10.METHODOLOGY OF IMPLEMENTATION :

For machine of Simhadri, tripping of generator is connected from second stage of rotor earth fault.The I.R. value of rotor is monitored and if it falls below 40 K-ohm, alarm and below 5 K-ohm, trip is initiated.

Alarm (<40 K-ohm) and trip (<5 K-ohm) values will be ensured by testing the relay and associated circuit. Before the unit is started, rotor earth fault relay operation and tripping will be ensured by shorting rotorwinding to earth at diode wheel ring / slip rings / brush holders.

11. FREQUENCY OF CHECKING AND MEASUREMENTS :

Once in during unit Overhaul.

12. RESPONSIBILITY CENTRE FOR IMPLEMENTATION :

Basic responsibility lies with respective Area in-charge i.e EMD-Lab group.

Overall responsibility lies with HOD(EMD)-I/C.

13. EXCEPTION REPORTING & DISPENSATION OF DEVIATION :

Any deviation shall be reported to HOD (EMD) and can be dispensed with his approval.

14. CHECKLIST FOR PREVENTION AND DETECTION OF GENERATOR ROTOR**EARTH FAULTS:**

Sl no	Activity	Frequency	Responsibility
1	Control of moisture in generator casing gas- Checking of dew point	Daily	Operation
2	Check the liquid level detectors periodically- Checking of LLD oil Checking of seal oil parameter periodically.	Daily	Operation
3	Prevention of dust entry to prevent contamination of rotor	During Generator OHor Whenever rotor is threaded out	EMD
4	The healthiness of generator rotor earth fault monitoring relay	Daily	EMD

15. RECORD KEEPING:

Records will be kept with Electrical Testing group and Generator group of Electrical Maintenance in files as mentioned earlier in this document.

16. REVIEW:

Documents will be reviewed once in two years by HOP.

